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C. A Beautiful Sorted List

time limit per test: 1 second

memory limit per test: 1024 megabytes

Forbidden Keywords for the Quiz: `sort`, `open`, `file`, `creat`(, `fstream`, `thread`, `process`, `system`(, `exec`(

Alice and Bob are two friends. Alice has a list of length N in **non-decreasing** order, and Bob has a list of length M , also in **non-decreasing** order.

Now, they want to combine their lists into a single non-decreasing list of length $N+M$. However, they are not very good at algorithms, so they asked for your help.

Since you are a computer science student, your task is to write an efficient algorithm to merge the two given lists into one non-decreasing list. **Solve the problem in $O(N+M)$.**

Input

The first line contains an integer N ($1 \leq N \leq 10^6$), denoting the length of Alice's list.

The second line contains N space-separated integers representing Alice's list.

The third line contains an integer M ($1 \leq M \leq 10^6$), denoting the length of Bob's list.

The fourth line contains M space-separated integers representing Bob's list.

All the numbers given in the input will fit within a **32-bit signed integer**. It is guaranteed that the given lists will be in **non-decreasing** order.

Output

You have to make a sorted list in **non-decreasing** order from the given lists and show the output.

Examples

input	Copy
<pre>4 1 3 5 7 4 2 2 4 8</pre>	
output	Copy
<pre>1 2 2 3 4 5 7 8</pre>	

input	Copy
<pre>3 2 10 12 6 3 4 6 7 8 9</pre>	
output	Copy
<pre>2 3 4 6 7 8 9 10 12</pre>	

input	Copy
<pre>5 1 2 3 4 5</pre>	

CSE221 Assignment 02 Summer 2026

Contest is running

42:35:51

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

2
10 12
output
1 2 3 4 5 10 12

input
4
1 2 12 13
3
10 15 18
output
1 2 10 12 13 15 18

input
8
1 2 3 8 8 10 12 14
9
1 1 4 5 6 8 13 15 16
output
1 1 1 2 3 4 5 6 8 8 8 10 12 13 14 15 16

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